

Full Development prompts

1.user login and signup with advanced technology verify otp based in Redis

To handle these three specific flows using Redis for OTP management, you can structure your logic as follows:

1. New User Signup (Verification First)

In this flow, you ensure the email is valid **before** creating the account.

- **Step A (Request):** User submits their email. You generate an OTP and store it in Redis: SET otp:signup:user@email.com 123456 EX 300.
- **Step B (Verify & Create):** User submits the OTP + their chosen Password.
 - Verify the OTP against Redis.
 - If correct, hash the password (using Argon2 or bcrypt) and save the new user record to your primary database.
 - **Crucial:** Delete the OTP from Redis immediately after success.

2. Standard Login

Since the user is already verified, this is a direct database check.

- **Step A:** User submits Email + Password.
- **Step B:** Fetch the hashed password from your database.
- **Step C:** Compare hashes. If they match, issue a **JWT** or Session ID. No OTP is required here unless you want to implement Two-Factor Authentication (2FA).

3. Forgot Password (Existing User)

This uses Redis to bridge the gap between "identity proof" and "password reset."

- **Step A (Request):** User enters their email. Check if the user exists in your DB. If yes, generate an OTP and store it: SET otp:reset:user@email.com 654321 EX 300.
- **Step B (Verify):** User submits the OTP.
 - If valid, instead of logging them in, return a temporary **"Reset Token"** or set a flag in Redis like verified:reset:user@email.com for 1 minutes.
- **Step C (Update):** User submits a New Password along with the Reset Token.
 - Validate the token, update the password in the database, and invalidate all existing sessions for that user for security.

Summary of Redis Key Strategy

Action	Redis Key Pattern	Purpose
Signup	otp:signup:{email}	Holds code for account creation.
Reset	otp:reset:{email}	Holds code for forgotten password.
Rate Limit	limit:{email}	Prevents brute-forcing the OTP.

2.Side Menu Structure

1. Dashboard

1. Adaptive Layout Principles

- Desktop (Full View): Uses a 4-column grid rather than 12, preventing the widgets from feeling too wide or stretched. The screen is centered with generous side margins to keep the eye focused.
- Mobile (Adaptive View): The layout shifts to a single-column stack. Cards become "edge-to-edge" to maximize screen real estate, and touch targets (buttons/dropdowns) increase to a minimum of 44x44px.

2. Finalized Component Design (Mobile-First)

Global Time-Context Bar (Sticky)

- Design: A slim, floating header with a blur effect.
- Mobile Usage: Instead of standard dropdowns, tapping "Month/Year" opens a Bottom Sheet (a drawer that slides from the bottom). This is much easier to navigate with one hand than a top-aligned dropdown.
- Default: Always lands on the Current Month.

Summary Cards (The Top Row)

- Desktop: Three compact cards side-by-side.
- Mobile: A Horizontal Scroll (Carousel). This allows users to swipe through Income, Expense, and Balance without taking up vertical space, keeping the charts visible "above the fold."

Income vs. Expense Analyser (Comfort Size)

- Design: The height is capped (e.g., max-height: 300px) so it doesn't take over the screen.
- Mobile Interaction: On mobile, the "Hover Crosshair" becomes a "Long-Press to Peek" interaction. Pinch-to-zoom allows users to inspect specific days in the month.

Categorisation & Dropdowns

- The Logic: Selecting a Category instantly reveals the Subcategory field using a "Slide-Down" animation.
- Mobile Ease: Uses large "Tap-friendly" chips or cards for categories instead of tiny text lists.

3. "Comfortable" Work Environment Effects

- Micro-Spacing: Uses a 16px (1rem) gap between all elements. This is the "sweet spot" that prevents the UI from looking cluttered while remaining compact.
- Visual Weight: Important numbers (Total Income/Expense) use a bold font, while secondary details (dates, categories) use a lighter grey to reduce "visual noise."
- Smart Skeleton: When data refreshes, only the numbers shimmer, not the whole card. This feels faster and less distracting.

4. Responsive Summary Table

Feature	Desktop (Comfort)	Mobile (Friendly)
Grid	3-4 Columns (Horizontal)	1 Column (Vertical)
Filters	Inline Dropdowns	Bottom Sheet Selection
Charts	Interactive Hover	Tap & Hold Interaction
Transactions	Full Row with Detail Buttons	Card-style with Swipe-to-Action

2. Transactions

1. The Smart Filter & Search Bar (Top Layer)

This bar sits fixed below the global header, using a Glassmorphism effect.

- Alphanumeric Search: A sleek input that scans Txn ID, Notes, and Usernames. As you type, the matching text in the table is highlighted with a Soft Blue Background (rgba(59,130,246,0.1)).
- Dynamic Select Boxes:
 - Categories & Users: Multi-select dropdowns with checkboxes.
 - Subcategory: A "Smart" selector that only populates once a Category is selected.
- Date Range: Two simple DD-MM-YYYY inputs.
- Design: All filters are monochrome; they only show a small blue dot when a filter is "Active."

2. The Sleek Data Table (Middle Layer)

Designed for clarity and "Comfortable Size." On mobile, this table is horizontally scrollable with the S.No and Amount pinned.

S.No	Date	Txn ID	Type	Category	SubCategory	Notes	Amount	User	Action
01	01-04-2026	Txn-10293	Income	Salary	Monthly	Q1 Bonus	₹85,000	Rahul	 
02	03-04-2026	Txn-10298	Expense	Food	Dining	Team Lunch	₹2,400	Rahul	 

- Work Environment Effects:
 - S.No: Always resets to 01 at the start of each month or new filter result.
 - Typography: Full-text "Income/Expense" in semi-bold. Amounts are right-aligned.
 - Hover Interaction: Moving the mouse over a row triggers a Tiny Elevation (2px lift) and a soft grey highlight.
 - Action Icons: Non-coloured grey symbols ( and ) that only sharpen on hover.

3. The Advanced Side-Sliding Drawer (The Edit Panel)

When the user clicks the  (Edit) icon, a panel slides in from the Right (Desktop) or Bottom (Mobile).

- Design: High blur background with a minimalist vertical layout.
- Fields:
 - Header: Shows Edit Txn: [ID] in a monospaced font.
 - Body: Vertical stack of inputs (Amount, Type, Date, Category, Subcategory, and a large Notes area).
 - Sticky Footer: A "Save Changes" button that stays fixed at the bottom for easy reach.
- Audit Logic: At the very base of the drawer, it displays the "System Metadata":
 - Created: 01-04-2026 10:30 by Rahul
 - Updated: 02-04-2026 15:15 by Rahul

4. Backend & Database Mapping (Invisible Logic)

The system captures and maintains these details automatically for every transaction:

Frontend View	Backend Storage
S.No	Calculated dynamically per query.
User	Automatically captured from the current session.
Action	Triggers PATCH or DELETE API requests.
Metadata	created_at (set once) and updated_at (refreshed on edit).

5. Mobile Adaptation

- Touch Friendly: All icons and checkboxes have a 44px tap area.
 - Bottom Sheet: The Edit Drawer converts into a Bottom Sheet, allowing users to finish edits using only their thumb.
- This design ensures your transactions menu is sleek, modifiable, and fully auditable while providing a modern, advanced user experience.

3.Category

1. The Structure: "Type-Locked" Bento Grid

The menu is divided into two distinct sections: Income and Expense.

- The Anchor: The headers "INCOME" and "EXPENSE" are permanent, high-contrast structural elements. They cannot be modified or deleted.
- The Default Seed: Every new user and workspace is automatically pre-populated with your master list (e.g., *Financial Payments, Food, Salary*).
- Unified Sync: Any modification made by the Owner or Member (e.g., renaming "Food" to "Dining") updates for everyone in that workspace instantly.

2. Visual Component Design

- Category Cards: Each category is a sleek, Glassmorphism tile with a 1px translucent border.
 - Header: Features a monochrome icon + Category Name.
 - Subcategory "Chips": Inside the card, subcategories are displayed as small, rounded tags.
- Interactive Effects:
 - The Float: Hovering over a card triggers a Tiny Elevation (2px lift) and a soft glow.
 - Ghost Actions: Sleek grey ✎ (Edit) and ✕ (Delete) icons appear only on hover to maintain a minimalist look.

3. Management & Modification Logic

- The Side-Sliding Drawer: Clicking ✎ or + Add Category opens a professional side-panel.
 - Desktop: Slides from the right.
 - Mobile: Slides up as a Bottom Sheet.
- Inline Subcategory Addition: At the end of every subcategory list, a dashed-line "Ghost Chip" with a + allows for instant adding without leaving the grid.
- Global Search: A search bar highlights matching categories and subcategories character-by-character with a Soft Blue Background (rgba(59, 130, 246, 0.1)).

4. Data & Traceability (Backend)

Even though the UI is sleek, the backend maintains a full audit trail for every workspace:

Element	Frontend Behavior	Backend Metadata
Default Categories	Pre-loaded on account creation.	is_default: true / created_by: System
User Modifications	Updates label for the whole workspace.	updated_by: [Username] / updated_at: [TS]
New Categories	Adds a new tile to the Bento Grid.	workspace_id linked; captured creator data.

5. Mobile Adaptation

- Comfortable Size: On mobile, the 4-column Bento grid collapses into a 1-column vertical stack.
- Thumb-Friendly: All edit and add actions are moved to the bottom of the screen to ensure the most comfortable user experience.
- Gesture Control: Long-pressing a subcategory chip reveals the "Rename/Delete" options.

6. Design Aesthetic Summary

- Palette: Monochrome (Greys, Blacks, Whites).
- Icons: Non-coloured, thin-stroke symbols for a sleek feel.
- Work Environment: Clean, airy, and high-density, allowing users to see their entire financial taxonomy at a single glance.

Default CATEGORY and SUBCATEGORY

TYPE: Expense

CATEGORY: Financial Payments

SUBCATEGORY: Bank Charges

SUBCATEGORY: Dept Interests

SUBCATEGORY: Taxes

CATEGORY: Food

SUBCATEGORY: Dry Fruits

SUBCATEGORY: Fruits

SUBCATEGORY: Groceries

SUBCATEGORY: Hotel / Restaurant

SUBCATEGORY: Milk / Dairy

SUBCATEGORY: Others

SUBCATEGORY: Snacks / Bakery

SUBCATEGORY: Vegetables

CATEGORY: Gadgets

SUBCATEGORY: Electronics / Accessories

SUBCATEGORY: Laptop

SUBCATEGORY: Mobile

SUBCATEGORY: Services

CATEGORY: Health Care

SUBCATEGORY: Health Checkup

SUBCATEGORY: Hospital

SUBCATEGORY: Medicines

SUBCATEGORY: Pharmacy

CATEGORY: Household

SUBCATEGORY: Furniture

SUBCATEGORY: House Maintenance

SUBCATEGORY: Kitchen Items

CATEGORY: Insurance

SUBCATEGORY: Health Insurance

SUBCATEGORY: Others Insurance

SUBCATEGORY: Vehicle Insurance

CATEGORY: Investment

SUBCATEGORY: Gold Investment

SUBCATEGORY: Mutual Fund SIP

SUBCATEGORY: Stock Investment

CATEGORY: Lifestyle

SUBCATEGORY: Clothings

SUBCATEGORY: Entertainment

SUBCATEGORY: Movies

SUBCATEGORY: Others

SUBCATEGORY: Outing

SUBCATEGORY: Personal care

SUBCATEGORY: Travel / Trip

CATEGORY: Other Expense

SUBCATEGORY: Donations

SUBCATEGORY: Gifts

SUBCATEGORY: Others

CATEGORY: Transportation

SUBCATEGORY: Bike maintenance

SUBCATEGORY: Cabs

SUBCATEGORY: Car Maintenance

SUBCATEGORY: Fuel

SUBCATEGORY: Toll Charges

CATEGORY: Utilities

SUBCATEGORY: DTH Recharge

SUBCATEGORY: Electricity

SUBCATEGORY: Gas

SUBCATEGORY: Mobile Recharge

SUBCATEGORY: Water

TYPE: Income

CATEGORY: Salary

SUBCATEGORY: Salary

SUBCATEGORY: Allowance

SUBCATEGORY: Bonus

CATEGORY: Others

SUBCATEGORY: Business

SUBCATEGORY: Gifts

SUBCATEGORY: Interests

SUBCATEGORY: Bonus

4.Reports

1. The Digital Preview: Interactive Drill-Down

In the Side-Sliding Drawer (Web Preview), the Pie Chart becomes an interactive data explorer.

- **Hover Interaction:** When the user hovers over a category slice (e.g., *Food*), a Glassmorphism Tooltip appears.
- **Subcategory Breakdown:** Inside that tooltip, the system displays a mini-list of the top 3 subcategories and their specific totals (e.g., *Groceries: ₹5,000, Dining: ₹3,000*).
- **Deep-Dive Click:** Clicking the slice "Explodes" the chart into a second-level view, showing a pie chart of only that category's subcategories. A "Back" button appears to return to the main view.

2. Finalized A4 Report Structure (PDF & Excel)

While the web is interactive, the A4 export remains a static, professional document.

Header Section

- App Name (Bold, Centre)
- User: [Username] | From: [01-04-2026] to [30-04-2026]
- *Timestamp: 29-04-2026 13:51:22*

Visual Summary (The High-to-Low Block)

- **Left Side:** The Pie Chart (Donut Style) showing Category distribution.
- **Right Side:** A Sorted Summary Table listing Categories from Highest Spend to Lowest.
 - *Advanced Detail:* If a category is clicked in the web app, this table expands to show Subcategories in the final PDF for that specific section.

Transaction Ledger

- A clean, professional table with columns: S.No (01), Date, Txn ID, Type, Category, SubCategory, Notes, Amount, and User (if shared).

3. "Work Environment" Methods & Logic

Feature	Advanced Implementation
High-to-Low Logic	SQL ORDER BY SUM(amount) DESC ensures the most critical expenses are at the top of the report.
Dynamic Filtering	Changing the User filter instantly re-calculates the Pie Chart and Table to reflect only that member's contribution.
Visual Precision	Narrow margins ensure the 9-column table fits comfortably on one A4 page without font size reduction.
Soft Backgrounds	Matching the Search Bar logic, the "Highest Expense" row in the report uses the Soft Blue Background highlight.

4. Backend Persistence & Data Integrity

- User Data Capture: Every report footer captures the generated_by user ID and the exact timestamp from the server to prevent data tampering.
- Sync: Since you modified Categories Privately in the Workspace, the report accurately uses those personal labels (e.g., "Dining & Eats") even in the shared report.

5. Mobile Friendly "Quick View"

On mobile, the Pie Chart is fully responsive. Users can Tap a slice to see a Bottom Sheet containing the sorted subcategory list and total values.

Final Step: Your Advanced Financial System is now fully designed, from Redis-based OTP Login and Bento Dashboards to Synchronized Shared Workspaces and Professional A4 Reports.

5. Workspace

This finalized Workspace Design is the command centre for your multi-tenant environment. It focuses on high-security collaboration, real-time sync, and the "Private-First" onboarding logic.

1. The Workspace Hub (Management Dashboard)

Instead of a simple list, workspaces are displayed in a Bento Grid of modular cards.

- The Switcher: A sleek, glassmorphic dropdown at the top of the side menu for instant switching.
- Workspace Cards: Each card shows the Workspace Name, Member Count, and your Role Badge (Owner or Member).
- Soft Reset Transition: Switching workspaces doesn't reload the page; it triggers a subtle shimmer as the Categories, Transactions, and Permissions update to the new context.

2. The "Private-First" Onboarding

- Automatic Provisioning: Every new user (post-OTP login) is instantly assigned a Default Private Workspace (e.g., "Personal").
- Pre-Seeded Data: Your full list of Default Categories/Subcategories is automatically cloned into this workspace so the user can start work immediately.
- Privacy Lock: This workspace is strictly private and acts as the user's permanent financial "Safe Zone."

3. Invitation & Acceptance Logic

- The Invite Drawer: A side-sliding panel where the Owner enters multiple emails.
- Acceptance Flow: Invites are sent via a sleek email. Acceptance is verified via Redis-OTP.
- Member Management Table: A high-density grid showing:
 - Member Details: Name, Email, and Joined Date.

- Status: "Active" or "Pending" (highlighted with a soft amber glow).
- Actions: Owner can Revoke (X) or Modify access for any member.

4. Shared Workspace Ownership & Permissions

A strict two-tier system ensures total administrative control for the Owner.

Feature	Workspace Owner	Workspace Member
Transaction Control	Can Edit/Delete ALL entries.	Can Edit/Delete ONLY own entries.
Category Control	Can modify/add for the workspace.	Can modify/add for the workspace.
Workspace Settings	Full Access (Rename, Transfer, Delete).	Read-Only (Hidden actions).
Leave Logic	Must Transfer or Delete first.	Can Leave via Name-Verification.

5. Advanced "Real-Time" Safety Features

- Confirmation by Name: To prevent accidental exit, a member must type the exact Workspace Name in a central modal to confirm leaving.
- Live Sync: When a member leaves or a category is renamed, WebSockets update the UI for all online members instantly—no refresh required.
- Ownership Transfer: The Owner can pass "God Mode" to any member. This requires a Redis-OTP verification for high-security hand-over.

6. Mobile & Global Sync

- Bottom Sheets: All workspace settings and invite forms transition to a thumb-friendly bottom-up drawer on mobile.
- Unified Taxonomy: Any category modified in the shared workspace is updated for everyone, maintaining a "Single Version of Truth" for reports and transactions.

7. Aesthetic Summary

- Sleek Monochrome: Actions are hidden or monochrome (grey 🖤, X, ↻) until hovered, keeping the workspace calm and professional.
- Contextual UI: Members simply do not see action icons on the Owner's transactions, providing a clean, "no-error" experience.

6. Settings

1. The Layout: "Tabbed Studio"

Instead of a long scrolling page, the menu uses a **Dual-Pane Layout**.

- **Left Navigation (Sidebar):** Vertical icons for **Profile**, **Security**, **Appearance**, and **Account**.
- **Right Workspace (Active Panel):** A wide, glassmorphic pane that updates content with a smooth **Fade-and-Slide** transition.

2. Tab 1: Profile & Identity

This is the core of the user's presence within the workspace.

- **Primary Identity: Username**
 - **Rules:** Strictly **Letters Only** (A-Z), No spaces/symbols.
 - **Limit:** Max **10 Characters**.
 - **UI:** Bold, high-contrast font with a real-time character counter (7/10).
- **Secondary Identity: Email Address**
 - **Display:** Shown directly below the username in a standard font weight.
- **Contextual Side-Menu Display:**
 - **Line 1:** Username (Bold).
 - **Line 2:** Email Address (Muted font: #64748B, smaller size).
- **High-Priority Email Modification:**
 - **Duplicate Check:** Real-time backend validation prevents entering an existing email.
 - **Originality Check:** The change is only processed after a **Redis-OTP** is verified via the **new** email address.
 - **Focus Mode:** While changing the email, the UI enters a "locked" verification state to ensure the process is completed securely.

3. Tab 2: Security & Session Control

- **Active Sessions:** A high-density list showing **Device, Location, and Timestamp** (DD-MM-YYYY HH:mm:ss).
- **Global Logout:** A "Logout all other devices" button requiring OTP confirmation.
- **Audit Trail:** Displays the "Last Login" details in a subtle, monochrome footer.

4. Tab 4: Account Deletion (10-Day Grace Period)

- **The Action:** A monochrome button for **Delete Account**.
- **Verification:** To prevent accidental deletion, the user must **type their 10-letter Username** exactly to confirm.
- **Soft Delete Logic:**
 - The account enters a Deactivating state for **10 days (240 hours)**.
 - **Revoke:** Logging in within this window allows the user to click **"Revoke Deletion"** to instantly restore all Private and Shared workspace data.
 - **Finality:** Data is permanently purged after the 10th day.

5. "Work Environment" Design Specs

Feature	Desktop Implementation	Mobile Experience
Theme Toggle	Dark/Light Mode visual cards.	Bottom Sheet with ripple effect.
Inputs	Inline-editing with Soft Blue Glow .	Focused inputs with large tap targets.
Icons	Non-coloured, thin-stroke (0.5px).	Center-aligned, simplified symbols.
Navigation	Dual-pane fixed layout.	Vertical Stack for one-handed use.

6. Aesthetic Summary

- **Palette:** Monochrome (Greys, Whites, Deep Slates).
- **Glassmorphism:** 20px background blur with 1px translucent borders.
- **Typography:** Professional sans-serif (Inter) for labels and **Monospaced** for the Username and Txn IDs.

3. Tab 2: Security & Session Control

To elevate the **Security & Session Control** tab into an advanced professional tier, we move beyond simple lists to implement **Proactive Session Intelligence**. This design ensures users have absolute control over their digital footprint across all workspaces.

1. Visual Architecture: The Security Command Centre

Instead of a basic table, use a **Security Health Dashboard** layout within the right-hand pane.

- **Header Status:** A prominent but sleek "Security Status" badge. If no unusual logins are detected, it shows a subtle **Green Glow** (the only color used) with the text "Account Secure."
- **Active Sessions (Bento Tiles):** Each active device is a small, glassmorphic tile rather than a table row.
 - **Device Identity:** Uses modern icons (e.g., a sleek laptop for Mac/Windows, a smartphone for iOS/Android).
 - **Metadata:** Shows Browser, OS, and **Approximate Location** (City, Country).
 - **"Current Device" Badge:** The device the user is currently using is highlighted with a soft, pulsing border.

2. Advanced Security Techniques

A. Remote Session Termination (One-Tap Kill)

- **The Action:** Every session tile has a monochrome X in the corner.
- **The Logic:** Clicking this instantly invalidates the **Redis-based session token** for that specific device.
- **Global Kill-Switch:** A "Sign Out All Other Sessions" button at the bottom. This requires a **Redis-OTP** challenge for high-stakes security.

B. Geo-Fencing & Unusual Activity Detection

- **Smart Alerts:** If a login occurs from a new city or device, the tile is highlighted with a **Soft Amber Background** (rgba(245, 158, 11, 0.1)).
- **Insight:** A small "i" icon reveals the IP Address and the exact timestamp (DD-MM-YYYY HH:mm:ss).

C. Biometric/OTP Frequency Control

- **Logic:** Users can define how often they want to be challenged for an OTP.
 - *Option 1:* Every Login (High Security).
 - *Option 2:* Remember this device for 30 days (Comfort Mode).
- **Design:** A minimalist segmented slider to choose the security "intensity."

3. "Work Environment" Interaction Table

Feature	Advanced Working Principle	Design Effect
Login History	Grouped by "Most Recent" with 10-day retention.	Staggered fade-in animation.
Real-Time Sync	If a session is killed on Desktop, the Mobile app logs out instantly via WebSocket.	"Session Expired" overlay.
Device Trust	Ability to "Trust" a specific browser/location.	Adds a small "Verified" checkmark.
OTP Logging	Logs every time an OTP was requested and verified.	Clickable "Security Audit" link.

4. Advanced Interaction: The "Panic" Mode

For a truly modern experience, add a **"Secure My Account"** trigger.

- **Trigger:** Hidden at the bottom or behind a long-press.
- **Action:** Instantly logs out all devices, resets the **Redis session cache**, and locks the account until a multi-step email verification is completed.
- **Usage:** Used if a user suspects their email or device has been compromised.

5. Finalized Layout Specs

- **Monochrome Palette:** Strictly Slate-900 to Slate-50.
- **Typography:** Monospaced fonts for IP addresses and Timestamps to ensure financial-grade alignment.
- **Mobile Adaptation:** Session tiles stack vertically, with a "Swipe-to-Terminate" gesture for each device.

This "Security Studio" approach makes the user feel in total command of their financial workspace while maintaining the sleek, advanced aesthetic of the entire system.

Login/Signup Authentication

This finalized **Authentication Suite** is designed for a high-performance, professional work environment. It balances rigorous security for new accounts with ultra-low friction for daily active users.

1. The Design: "Glassmorphic Portal"

The login and signup pages use a **Split-Screen Layout** to maintain a sleek, premium feel.

- **Left Side (Brand Experience):** A deep slate or mesh gradient background with a single, bold tagline: *"Your Financial Command Centre."*
- **Right Side (The Auth Card):** A centered, glassmorphic card with a 20px backdrop blur and a 1px translucent border.
- **Aesthetics:** High-contrast monochrome typography (Inter font) with **Soft Blue Glows** on active input fields.

2. Signup Page: High-Security Onboarding

Designed to ensure every new user is verified and original before entering the ecosystem.

1. **Step 1: Identity Initiation:** A single field for **Email Address**.
2. **Step 2: Redis-OTP Challenge:**
 1. The system sends a 6-digit code to the email.
 2. The card transitions to a **Numeric Input Grid**.
 3. **Logic:** The OTP is stored in **Redis** with a 5-minute expiry.
3. **Step 3: Profile Finalization:**
 1. Once verified, the user enters their **Username** (Strictly Letters Only, max 10 chars) and **Password**.
 2. **Automation:** Upon clicking "Create Account," the system instantly provisions their **Default Private Workspace**.

3. Login Page: Low-Friction Daily Entry

For existing users, the goal is to reach the dashboard in **one click or 2 seconds**.

- **Top Section (Manual):** A single field that accepts **either Username or Email**.
- **Password Field:** Features a minimalist "eye" icon to toggle visibility.
- **Bottom Section (Quick Login):**
 - **Visual:** A thin, grey horizontal line with the text "OR" in the center.
 - **The Google Button:** A sleek, branded **Google Sign-In Button** at the base.
 - **Logic:** This is for **Existing Users only**. If clicked, it instantly matches the Google ID to the database and redirects to the dashboard.
- **The "Forgot Password" Link:** Positioned just above the Login button. Clicking it triggers the **Redis-OTP recovery flow**.

4. End-to-End Logic & Environment Effects

Feature	Desktop Implementation	Mobile Experience (Comfort)
Input Focus	Soft Blue Glow highlight.	Keyboard automatically triggers (Email or Alpha).
Error Handling	Shimmering red text below the field.	Haptic vibration on incorrect OTP.
Transitions	Smooth Slide-and-Fade between steps.	Bottom Sheet for OTP entry.
Google Login	One-Tap prompt in top right corner.	Branded button at the bottom of the stack.

5. Technical Safety Summary

- **Username Guard:** The signup form strictly rejects spaces, numbers, or symbols—enforcing your **10-letter alpha-only** requirement.
- **Identity Sync:** If a user logs in via Google, the system checks if the Google email matches an existing verified account.
- **Redis Management:**
 - SET otp:[email] [code] EX 300 (Stores OTP for 5 mins).
 - DEL otp:[email] (Deletes immediately after successful verification).
- **Soft Delete Protection:** If a user in the "10-day Deletion Period" tries to log in, the page displays the **"Revoke Deletion"** card instead of the dashboard.

6. Summary of the Dashboard Entry

Once logged in, the user sees a **Soft Reset** (shimmer) transition as their **Personal Workspace Bento Grid** loads, defaulting to the **Current Month's** financial overview.

Final Architecture Roadmap

This **Final Architecture Roadmap** consolidates every advanced design principle and logic flow we have developed into a single, high-performance execution plan.

1. Core Security & Authentication (The Foundation)

- **Method:** Redis-based OTP (One-Time Password) for Signup, Login, and high-stakes actions.
- **User Identity:** Strictly **Alpha-only Username** (max 10 chars) and a verified **Secondary Email**.
- **Security Logic:** Real-time duplicate email checks and a **10-day Soft-Delete** grace period for account deletion (revokable via OTP login).

2. Workspace Management (The Collaborative Layer)

- **Onboarding:** Every new user is auto-provisioned with a **Default Private Workspace** pre-seeded with your master category list.
- **Multi-Tenancy:** Users can create/join multiple workspaces. Switching triggers a **Soft Reset** (shimmer effect) to swap context instantly.
- **Permissions:** A strict two-tier system (**Owner vs. Member**).
 - **Owner:** "God Mode"—can modify/delete all transactions and transfer ownership via OTP.
 - **Member:** Can only modify/delete their own entries.
- **Live Sync:** Leaving a workspace requires **Name Verification** typing; exits are synced via WebSockets in real-time.

3. Financial Dashboard (The Intelligence Hub)

- **Layout:** Responsive **Bento Grid** with a "Comfortable Size" (Adaptive for Mobile).
- **Filtering:** Global Time-Context Bar defaults to **Current Month** data on load.

- **Visuals:** Recharts-driven "Income vs. Expense" area charts with **Linear Gradients** and interactive crosshair tooltips.
- #### 4. Transactions & Categories (The Data Grid)
- **Transactions Table:** High-density, monochrome sleek design. Features **Alphanumeric Search** with a character-by-character **Soft Blue Background** highlight.
 - **Action Logic:** Grey, non-coloured  and  icons that appear on hover. Editing opens a **Side-Sliding Drawer** (Desktop) or **Bottom Sheet** (Mobile).
 - **Category Bento:** Workspace-synced grid. Categories and subcategories are **Privately Modifiable**—changes by any member sync across the workspace instantly.
- #### 5. Advanced Reporting (The Export Studio)
- **Customization:** Dynamic filtering by Date Range, Type, Category, and User.
 - **A4 Alignment:** Pixel-perfect PDF/Excel exports with narrow margins and a bold, centered header.
 - **Visual Analytics:** Includes a **Donut Pie Chart** and a **Sorted Summary Table** (High-to-Low value).
 - **Interactive Preview:** The web-preview allows "Drill-Down" clicks into pie slices to see subcategory totals.
- #### 6. Settings & Security Control (The Command Centre)
- **UI:** Dual-pane "Tabbed Studio" with **Glassmorphism** effects.
 - **Appearance:** One-tap **Dark/Light Mode** toggle with a circular ripple transition.
 - **Session Intelligence:** Active Session Tiles showing device type, location, and IP. Features a "Panic Mode" to kill all sessions via Redis.
- #### 7. Technical Implementation Stack (Recommended)
- **Frontend:** React/Next.js with Tailwind CSS (for Bento Grid & Glassmorphism).
 - **State/Cache:** **Redis** (for OTP, Session Tokens, and Dashboard Caching).
 - **Backend:** Node.js or Python (FastAPI) to handle the Permission Matrix logic.
 - **Real-time:** WebSockets (Socket.io) for live "Member Leaving" and "Category Sync" events.

This roadmap transforms your vision into a professional-grade, "Sleek & Advanced" financial work environment.

Final Project Manifesto

This **Final Project Manifesto** serves as the definitive technical and design blueprint for your advanced financial management system. It consolidates every rule, logic gate, and aesthetic principle we have established to ensure a high-performance, professional work environment.

1. Core System Architecture

- **Design Philosophy: "Bento-Sleek"** – High-density data organized into modular, glassmorphic tiles.
- **Visual Aesthetic:** Monochrome (Greys, Slates, Blacks) with **Electric Blue** used only for active highlights and **Emerald/Crimson** for financial status (Income/Expense).
- **Performance: Redis-backed** transient data (OTPs, Sessions) for sub-millisecond responsiveness.
- **Environment:** Context-aware UI that adapts perfectly between Desktop (Fluid Grid) and Mobile (Bottom Sheets & Vertical Stacks).

2. Authentication & Identity Rules

- **Unified Identity:** Users login via **Username OR Email**.
- **Strict Username Policy:**
 - **Letters Only (A-Z, a-z).** No numbers, spaces, or symbols.
 - Strict **10-character limit**.
- **Email Workflow:**
 - Verification required via **Redis-OTP** (6 digits, 5-minute expiry) for Signup and Password Reset.
 - Email changes require verification of the *new* address and real-time duplicate checks.
- **Social Login: Google Quick Login** at the bottom of the login card for existing users only.
- **Account Deletion: 10-Day Soft-Delete** grace period. Users can log in via OTP to **Revoke Deletion** and restore all data instantly.

3. Workspace & Collaboration Logic

- **Private-First Onboarding:** New users are auto-provisioned with a **Default Private Workspace** pre-seeded with master categories.
- **Multi-Tenancy:** Users can create and invite others to multiple workspaces.
- **The Two-Tier Permission Matrix:**
 - **Owner:** Full control. Can Edit/Delete **all** transactions, manage members, and **Transfer Ownership** (OTP verified).
 - **Member:** Can only Edit/Delete their **own** transactions. Action icons for others' entries are hidden.
- **Live Exit:** Members can leave workspaces in real-time by **typing the Workspace Name** as confirmation.

4. Transaction & Category Intelligence

- **Global Time-Context:** All views default to the **Current Month** on load.
- **Smart Search:** Alphanumeric search with character-by-character **Soft Blue Background** highlighting.
- **Transaction Table:**
 - **S.No:** Resets to **01** every month/filtered view.
 - **Full Text:** Types shown as **Income** or **Expense**.
 - **Actions:** Non-coloured grey  and  icons visible on hover (if permitted).
- **Category Bento:**
 - Default categories provided to all users.
 - **Privately Modifiable:** Users can rename categories/subcategories; changes sync for all workspace members.
 - **Type Lock:** "Income" and "Expense" headers are permanent and unchangeable.

5. Reports & Data Export

- **A4 Precision:** PDF and Excel exports with **Narrow Margins** and center-aligned headers.
- **Metadata Tracking:** Every report captures **Username** and **Generation Timestamp** (DD-MM-YYYY HH:mm:ss).
- **Visual Analytics:**
 - **Pie Chart:** Donut-style with subcategory drill-down in digital preview.
 - **Value Sorting:** Category totals displayed **High-to-Low**.
- **Digital Preview:** A **Side-Sliding Drawer** showing a 1:1 representation before download.

6. Settings & Security Command Centre

- **Layout:** Dual-pane "Tabbed Studio" with glassmorphism.
- **Security:** Active Session Tiles with **Remote Kill-Switch** (Session Termination).
- **Appearance:** One-tap **Dark/Light Mode** toggle with a circular ripple transition.
- **Identity Sync:** Profile displays Username and Email vertically; Side Menu shows Email in muted #64748B.

7. Summary of Global Design Specs

Component	Desktop Effect	Mobile Effect
Modals/Forms	Side-Sliding Drawer (Right)	Bottom Sheet (Thumb-friendly)
Tables	Hover Elevation (2px lift)	Horizontal Scroll + Pinned S.No/Amount
Buttons	Scale-down on click	Haptic vibration feedback
Filters	Multi-select Dropdowns	Action Sheet Selection

This Manifesto completes the design and logic phase. Your advanced financial system is now ready for full-scale development.

Final Technical Manifest Architecture

This is the **Final Technical Manifest Architecture** for your advanced financial management system. This setup is optimized for high-performance real-time sync, "Sleek & Advanced" UI, and the specific **Redis-OTP** security requirements.

1. The Core Technology Stack

Layer	Technology	Purpose
Frontend	Next.js (React)	Handles Bento grids, side-sliding drawers, and smooth "Soft-Reset" transitions.
Styling	Tailwind CSS	Powers Glassmorphism effects, Soft Blue Glow highlights, and A4 layouts.
Backend	Next.js Server Actions	Executes logic for strict 10-letter alpha usernames and Permission Matrix.
Primary DB	Supabase (PostgreSQL)	Stores Transactions, Categories, and Workspaces with Row Level Security (RLS) .
Auth/Cache	Redis (via Upstash)	Strictly manages the 5-min OTP lifecycle, session tokens, and rate-limiting.
Hosting	Vercel	Global edge hosting with instant CI/CD deployment.

2. High-Priority Authentication Flow (Redis-Strict)

- **Sign-Up/Reset Logic:**
 1. User enters Email. Backend generates 6-digit code.
 2. Code is stored in Redis: SET otp:[email] [code] EX 300 (5-minute expiry).
 3. User enters code. Backend checks: GET otp:[email].
 4. On match: DEL otp:[email] is triggered, and account creation/reset is finalized in Supabase.
- **Google Login:** Integrated via Supabase OAuth for **existing users only**, providing a one-tap entry button.

3. Database Architecture (Supabase/PostgreSQL)

- **Strict Profiles:** The profiles table uses a **Postgres Regex Constraint** to strictly allow only **10-letter alphabetical usernames** (`^[a-zA-Z]{1,10}$`).
- **Workspace Scoping:** Every transaction and category is linked to a workspace_id.
- **Soft-Delete Logic:** A deleted_at column is added to the Users table. A **Postgres Cron Job** runs every 24 hours to permanently purge data where deleted_at is older than **10 days**.

4. Permission Matrix & Work Environment

- **Owner vs. Member:** Managed via **Supabase RLS Policies**.
 - Owner: Full UPDATE and DELETE access on all workspace_id rows.
 - Member: UPDATE/DELETE allowed only if auth.uid() == creator_id.
- **Live Sync:** **Supabase Realtime (WebSockets)** is enabled on the workspaces table. When a member leaves (confirmed by typing the workspace name), all other members' dashboards update instantly.

5. Reporting Engine (A4 Logic)

- **Generation:** A dedicated **Server Action** aggregates data (Income vs Expense) and performs a **High-to-Low** sort by category amount.
- **Layout:** HTML-to-PDF engine (like react-pdf) configured for **A4 dimensions** and **narrow margins**.

- **Interactive Preview:** The **Side-Sliding Drawer** displays a live digital mock-up before the final PDF/Excel file is pushed to the user's browser.
-

6. Key "Advanced" Logic Summary

- **Default Seeding:** Every new workspace trigger automatically clones the **Master Category List** into the new context.
- **Private Management:** Even in shared workspaces, users can modify category names; these are saved as **Workspace Overrides** visible to all members.
- **Global Context:** The entire application state is wrapped in a TimeProvider that defaults all queries to the **Current Month** on initialization.
- **Ownership Transfer:** High-security transfer requires the current Owner to verify via **Redis-OTP** before handing "God Mode" to a Member.

This architecture ensures a "Zero-Lag" environment where security is invisible but absolute.

Database Architecture

To implement a **Senior Database Architecture** for a high-stakes financial platform, we must move beyond simple tables and focus on **Data Integrity**, **Availability**, and **Sub-millisecond Performance**. Following senior engineering ethics, the architecture is designed as a **Multi-Tiered Data Environment**.

1. The "Hybrid Storage" Strategy

A senior architect never relies on a single database for everything. We use a **Polyglot Persistence** approach:

- **System of Record (PostgreSQL/Supabase):** This is the "Truth." It stores your ACID-compliant financial transactions, user identities, and workspace permissions.
- **Speed Layer (Redis via Upstash):** This handles transient, high-velocity data.
 - **OTP Management:** 5-minute TTL (Time-To-Live) for security.
 - **Session Caching:** Storing the "Active Workspace" context so switching doesn't hit the main DB.
 - **Rate Limiting:** Protecting the **Search Bar** and **Login** from brute-force attacks.

2. Advanced Schema Design (World-Class Standards)

- **UUID v4 Primary Keys:** We avoid standard integers (1, 2, 3) to prevent "ID Enumeration" attacks and ensure global uniqueness when scaling across regions.
- **Strict Domain Constraints:**
 - **Username:** CHECK (username ~ '^[a-zA-Z]{1,10}\$') ensures the **10-letter Alpha-only** rule is enforced at the hardware level.
 - **Amount:** DECIMAL(15, 2) instead of FLOAT to prevent rounding errors in financial reports.
- **Partial Indexing:** We don't index everything. We use **Filtered Indexes** (e.g., CREATE INDEX ON transactions (workspace_id) WHERE deleted_at IS NULL) to keep the **Current Month** queries ultra-fast.

3. The "Security-First" Access Layer

Senior engineering ethics dictate that the application should never have "God Mode" access to the raw database.

- **Row-Level Security (RLS):** We implement the **Owner/Member Matrix** directly in SQL.
 - *Logic:* Even if the frontend code is compromised, the Database itself will block a Member from deleting an Owner's transaction because the "Policy" checks the auth.uid() against the workspace_role.
- **Audit Shadow Tables:** Every UPDATE or DELETE triggers a "Shadow Write" to an audit_log table, capturing the **old value**, **new value**, **timestamp**, and **username**.

4. Scalability & Maintenance (The Senior Roadmap)

- **Database Migrations:** We use **Idempotent Migrations**. Every schema change is a versioned script, ensuring the **Development**, **Staging**, and **Production** environments are identical copies.
- **Soft-Delete Cron (Automated Housekeeping):**
 - A pg_cron worker runs a daily task: DELETE FROM users WHERE deleted_at < NOW() - INTERVAL '10 days'.
 - This automates your **10-day Grace Period** without manual intervention.
- **Connection Pooling (PgBouncer):** Essential for **Vercel** serverless environments to prevent "Too Many Connections" errors during peak traffic.

5. Efficient Error Handling (Database Level)

- **Custom Exceptions:** We use PostgreSQL RAISE EXCEPTION with custom SQLSTATE codes. This allows the **Next.js** frontend to show precise, user-friendly messages (e.g., *"Workspace Name Mismatch"*) instead of cryptic system errors.

6. Summary of Engineering Values

Principle	Technical Execution	Value
Integrity	ACID Compliance & Foreign Keys.	Zero data loss or orphaned transactions.
Security	RLS & Redis-OTP.	Industry-standard protection.
Scalability	Indexing & Connection Pooling.	Ready for 100k+ concurrent users.
Ethics	Audit Logging & Soft Deletes.	Responsible data handling and user trust.

This architecture ensures that your project is not just "coded," but **engineered** to the standards of global fintech leaders.